

Stall Detect Tuning Guide

Power Application Controller®



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1 INTRODUCTION

This document describes the procedure for tuning the PAC5XXX Stall Detection. The Stall implementation can be used for BLDC motor. The motor BEMF constant (V/Hz) is necessary for the Stall algorithm to work but it can also be estimated using this Stall tuning procedure.

The Stall algorithm calculates the motor BEMF voltage drive vector and compares it with the motor voltage drive vector. Stall detection accuracy may also depend on the Stall Time Out. Stall tuning needs a working motor setup and GUI with tuned XML file. Make sure to disable the IPD/SIM and re-enable IPD/SIM after Stall tuning is completed.

2 STALL TUNING OVERVIEW

During Stall tuning the following steps should be followed:

- Estimate BEMF Constant (V/Hz)
- Adjust Stall Time Out in seconds

3 STALL TUNING PARAMETERS

The following #defines in param_user.h can be used to store the Stall parameters after tuning with the GUI for operation without the GUI:

```
#define PUSER_VALUE_STALL_KV 0.0000f
#define PUSER_VALUE_STALL_WINDOW_SEC 0.00f
```

The following #defines in config_features.h and config_debug.h need to be un-comment for Stall Detection tuning process

```
#define ENABLE_STALL_DETECT
#define ENABLE_DEBUG_PWM DAC
#define ENABLE_DEBUG_DISPLAY
```

To tune Stall, user can adjust Estimated BEMF Constant and Stall Time Out from the GUI.

4 STALL TUNING PROCEDURE

The following Stall tuning procedure assumes that the motor FOC parameters have been tuned for the proper operation without Stall. See Fig 1 for initial Stall Detection Setup.

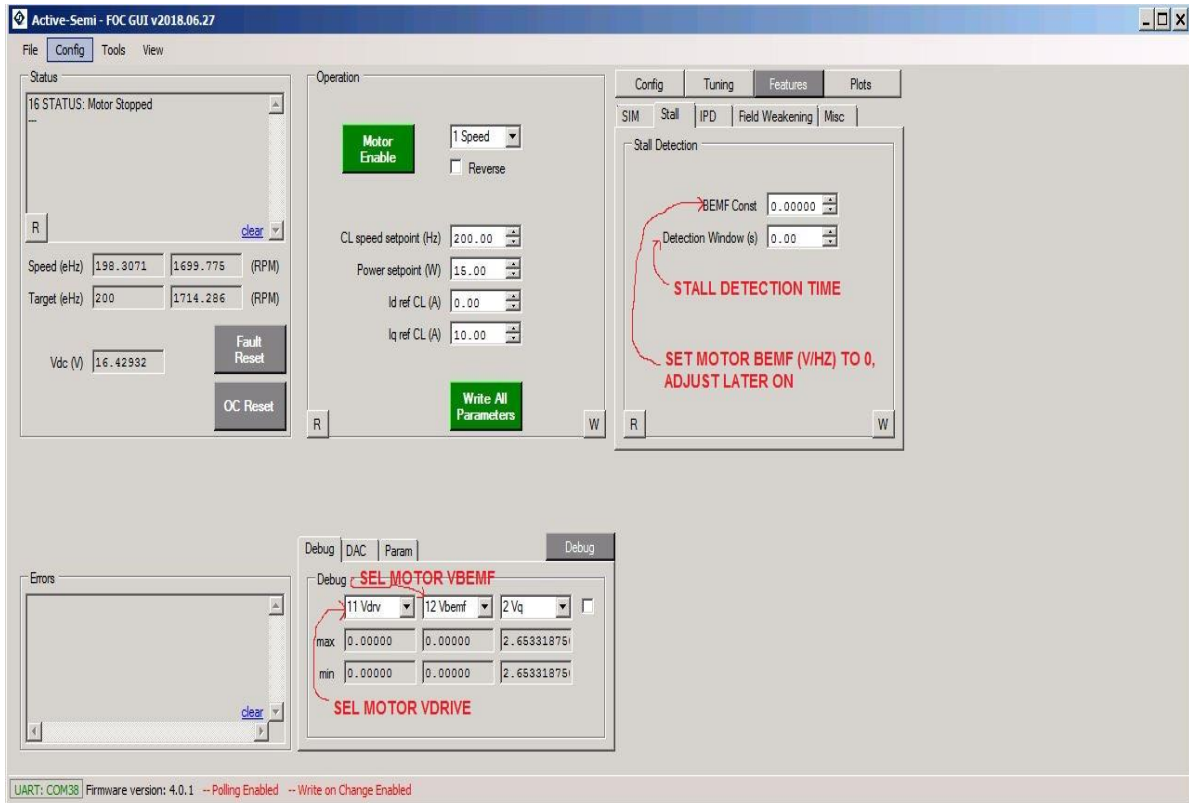


Figure 1: Stall GUI Setup

4.1 Set Stall Parameters

The BEMF Constant (V/Hz) is required to tune Stall along with the Stall TimeOut. If the motor BEMF Constant (V/Hz) is not available, then the user can use Debug variable Item index 11 and index 12 along with PWM DAC selection index 19 & 20 outputs to calculate an approximate value. Fig 2 & 3 shows how the GUI is given access to read and display Stall variables using DEBUG and PWM DAC options.

Fig 2 & 3 describes how to select Stall Detection variables using Debug and PWM DAC Output using GUI.



Figure 2: Select Stall variables using Debug

4.2 Estimate BEMF Constant

This section describes the procedure to verify Stall Detection.

1. Make sure motor is secure.
2. Observe the motor phase current using current probe.
3. Observe PWM DAC CH1 (Vdrive) and PWM DAC CH2 (Vbemf) outputs using voltage probe.
4. Disable IPD.
5. Disable SIM.
6. Un-comment ENABLE_STALL_DETECT in config_features.h.
7. Compile and download the code.
8. Power cycle the board.
9. Connect the GUI.
10. Load tuned and tested appropriate xml file.
11. Under Stall Tab, set BEMF Constant (V/Hz) = 0.000 and Stall TimeOut = 0.5 seconds, check Fig 1 for more information.
12. Select PWM DAC output to 19 Vdrive and 20 Vbemf under DAC and set gains to 0, check Fig 3 for further details.
13. Select Debug index 11 Vdrv (Vdrive) and Debug index 20 Vbemf (VBEMF).
14. Set closed-loop speed set point higher than open loop speed set point.



Figure 3: PWM DAC gain selection for Vdrv and Vbemf

15. Enable the motor.
16. Wait until motor switches to closed-loop mode and reaches the target closed loop speed.
17. Set motor closed-loop speed to the maximum speed you want to run the motor at, make sure the supply has enough current to run the motor up to the max speed.
18. Wait until the motor reaches the closed-loop speed target.
19. Display Vdrive, Vbemf by selecting Debug index 11 Vdrv and Debug index 12 Vbemf in the GUI Debug Tab. Check Fig 4 for further details.



Figure 4: Display Vbemf, Vdrive using Debug

20. Display Vdrive and Vbemf on the scope by selecting PWM DAC output index 19 and index 20 . Check Fig 5 for more information.



Figure 5: Display Vdrive (DAC CH1), Vbemf (DAC CH2) using PWM DAC Outputs

21. In the DAC tab Fig 6, set the appropriate DAC gains until Vdrv and Vbemf are visible on the scope (see Fig 9).



Figure 6: PWM DAC Gain Adjustment

22. Slowly raise the BEMF Constant (V/Hz). Keep the Stall time to 0.50 seconds for slow speed and 0.05 to 0.1 seconds for high speed applications (see Fig 7).

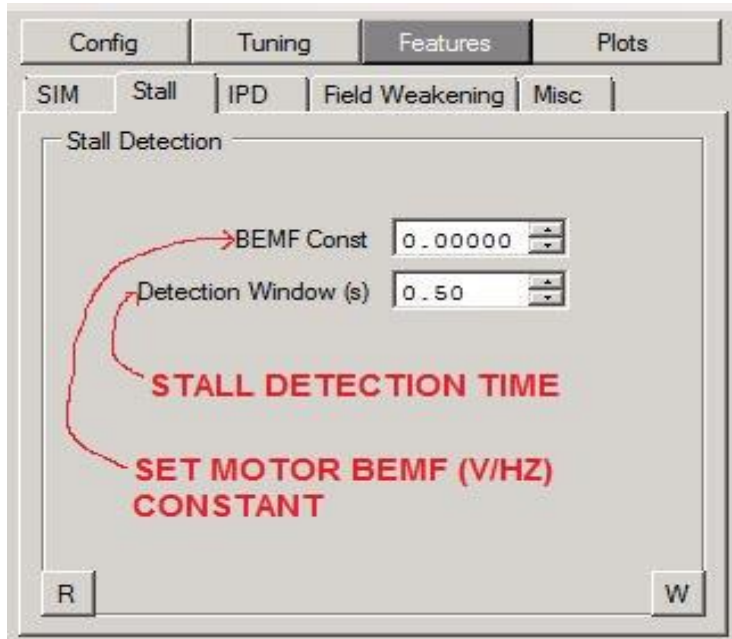


Figure 7: Set BEMF (V/Hz) Gain and Stall TimeOut

23. Repeat step 23 until Vbemf (Debug index 12) becomes larger than Vdrv (Debug index 11) (see Fig 8).

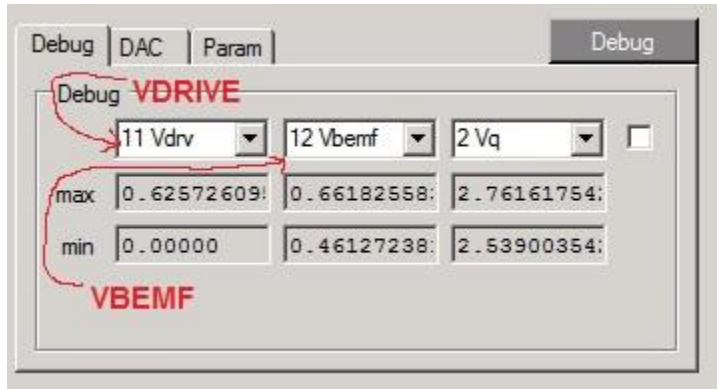


Figure 8: Display Vdrive, Vbemf using Debug

24. Fig 9 explains the complete Stall Detection flow using PWM DAC outputs. In closed-loop mode, when the motor Iq current becomes positive and the motor drive voltage (yellow line scope CH1) becomes less than vbemf (blue line CH2), the system enters in Stall timeout condition. If the condition remains true and the Stall time out expires, then system will post a Stall event, and the motor will stop as shown in Fig 9.

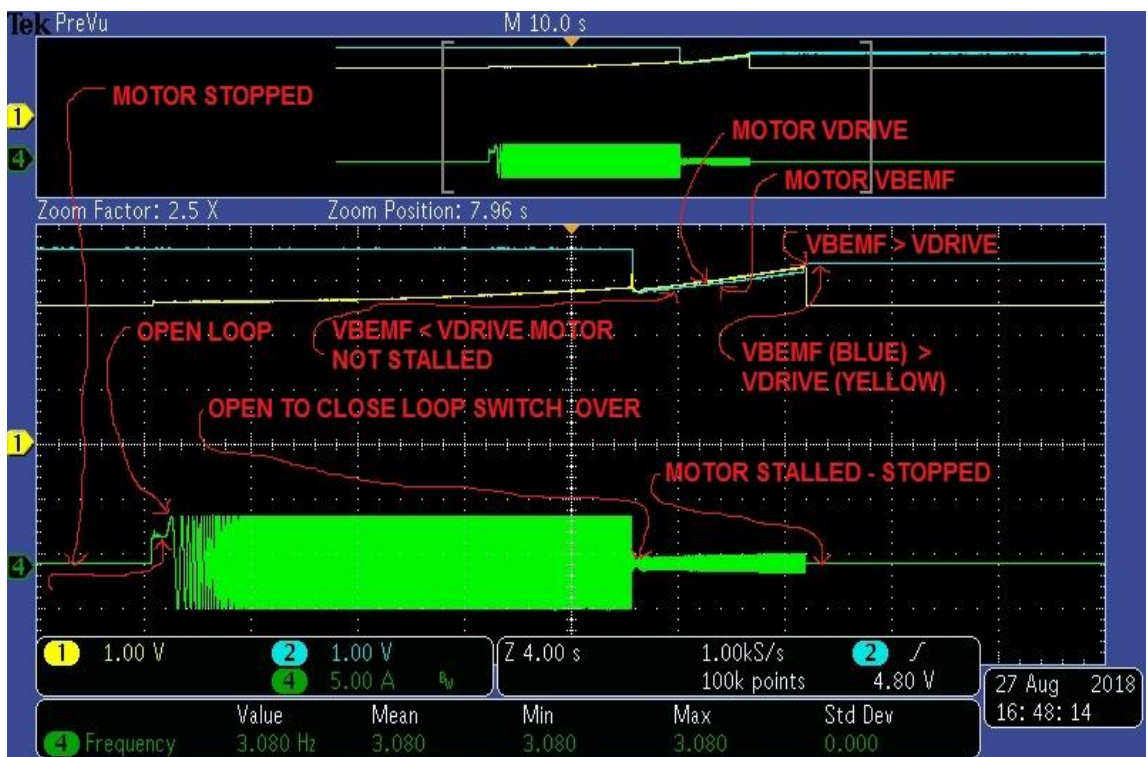


Figure 9: Display Vdrive (DAC CH1), Vbemf (DAC CH2) using PWM DAC Outputs.

Record the value of GUI BEMF Constant (V/Hz) entered into the Stall Tab. Use the final value 25% less than the GUI computes for the BEMF motor constant (see Fig 10).

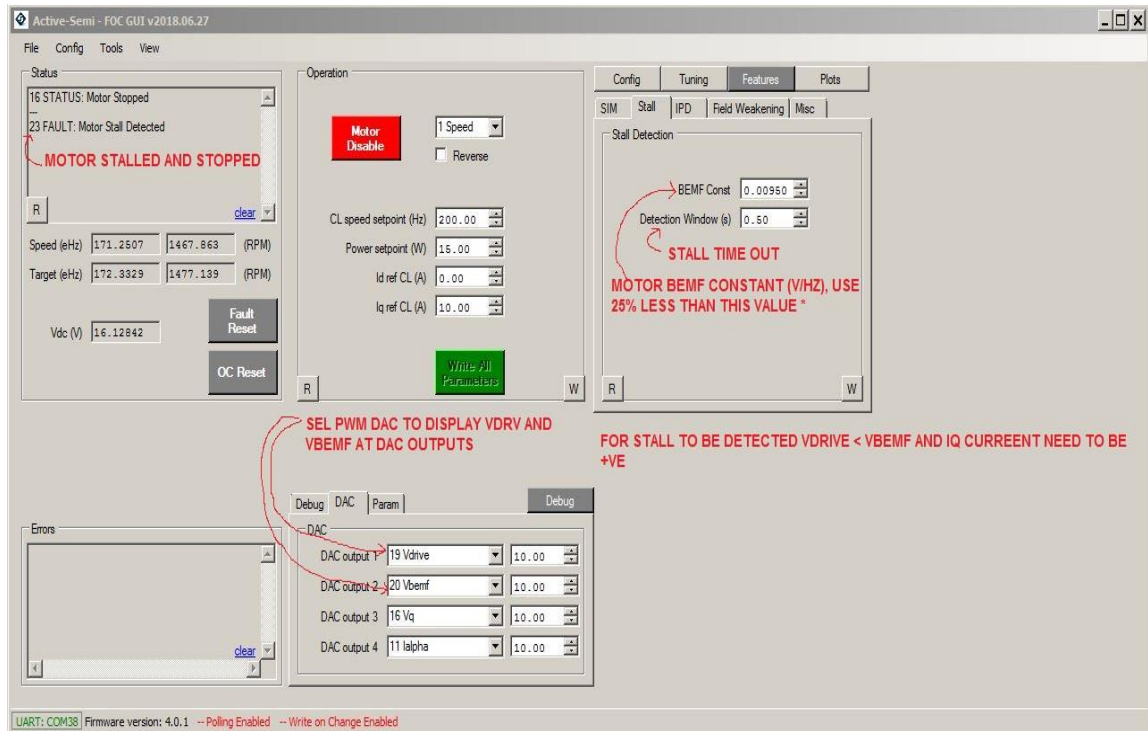


Figure 10: Use computed BEMF Constant (V/Hz)

25. Set the motor closed-loop speed to the minimum speed. Make sure that the speed is higher than the OL speed set point.
26. Use the last computed BEMF Const (V/Hz) from step 24.
27. Enable the motor.
28. Wait until the motor reaches the minimum closed loop speed,
29. Observe the motor. Check Vdrive and Vbemf.
30. Make sure Vdrive is more than Vbemf, otherwise, the motor will experience a false Stall.
31. Disable the motor.

4.3 Adjusting Stall Timeout

Generally, a Stall time out of 0.50 seconds for slow speed and 0.05 to 0.1 seconds for high speed applications should work well. If with these values, the controller reports a false Stall, then this value needs to be increased. See Fig 7 for more details.



CONTACT INFORMATION

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